Open Quantum Systems

Basit Reyaz

September 2025

1 Mathematical Formalism

Quantum mechanics is described by elements of the Hilbert space. Each physical system has an associated Hilbert space.
Definition. A *Hilbert space* $\mathcal{H}$ is a complete inner product space. That is, $\mathcal{H}$ is a vector space over $\mathbb{R}$ or $\mathbb{C}$ equipped with an inner product

$$\langle \cdot, \cdot \rangle : \mathcal{H} \times \mathcal{H} \to \mathbb{R} \text{ or } \mathbb{C}$$

satisfying:

(1) Conjugate symmetry:	$\langle x, y \rangle = \overline{\langle y, x \rangle}$,
(2) Linearity in the first argument:	$\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$,
(3) Positive-definiteness:	$\langle x, x \rangle \geq 0$ and $\langle x, x \rangle = 0 \iff x = 0$.

The inner product induces a norm $\|x\| = \sqrt{\langle x, x \rangle}$, and $\mathcal{H}$ is required to be *complete* with respect to this norm; that is, every Cauchy sequence in $\mathcal{H}$ converges to an element of $\mathcal{H}$.

A pure state ψ is an element of the Hilbert space. An observable acts as a self adjoint operator on $\mathcal{H}$. The operator can be expressed as a matrix, when $\mathcal{H} = \mathbb{C}^d$. The set of all bounded linear operators on $\mathcal{H}$ is denoted by $\mathcal{B}(\mathcal{H})$.
In quantum mechanics, physical observables are represented by *linear operators* acting on a Hilbert space $\mathcal{H}$ of state vectors.

A map $A : \mathcal{H} \to \mathcal{H}$ is called a *linear operator* if for all $|\psi\rangle, |\phi\rangle \in \mathcal{H}$ and scalars α, β,

$$A(\alpha|\psi\rangle + \beta|\phi\rangle) = \alpha A|\psi\rangle + \beta A|\phi\rangle.$$

An operator A is Hermitian or self adjoint if

$$\langle \psi | A\phi \rangle = \langle A\psi | \phi \rangle \quad \text{for all } |\psi\rangle, |\phi\rangle.$$

Important!!: Hermitian operators represent physical observables; their eigenvalues are real, and eigenvectors corresponding to different eigenvalues are orthogonal.

A reminder on eigenvalues and eigenvectors, measuring an observable A yields an eigenvalue λ and associated eigenstate $|\psi\rangle$ satisfying

$$A|\psi\rangle = \lambda|\psi\rangle.$$

The eigenvalues and eigenvectors can be found using,

$$\det(A - \lambda I) = 0 \quad (A - \lambda I)|\psi\rangle = 0$$

Spectral Theorem: A hermitian operator A can be expressed in terms of its eigenvalues and eigenvectors. If the set of eigenvectors is $\{|n\rangle\}$ and eigenvalues are a_n, then,

$$A = \sum_n a_n |n\rangle\langle n|$$

where $|n\rangle\langle n|$ is the projection operator P_n onto the eigen state $|n\rangle$. These projectors satisfy

$$P_n^\dagger = P_n \quad \sum_n P_n = I \quad P_n^2 = P_n \quad P_n P_m = \delta_{mn} P_n$$

Consider the example, taking an operator H.

$$H = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Eigenvalues:$\lambda_1 = 3,\ \lambda_2 = 1$.
Orthonormal eigenvectors:

$$|v_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad |v_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Spectral decomposition gives us,

$$H = 3\,|v_1\rangle\langle v_1| + 1\,|v_2\rangle\langle v_2|.$$

Thus, in the basis of the eigenvectors, each matrix can be represented as a diagonal matrix consisting of eigenvalues.
The same could be done through routine diagonalization process, For a matrix A, suppose it has (orthonormal) eigenvectors

$$v_1,\ v_2,\ \ldots,\ v_n.$$

We construct the matrix U by placing these eigenvectors as its columns:

$$U = (\, v_1\ v_2\ \cdots\ v_n\,).$$

If the eigenvectors form an orthonormal basis which happens only if $AA^\dagger = A^\dagger A$(A is normal) or $A = A^\dagger$(A is hermitian), then U is unitary:

$$U^\dagger U = I.$$

The diagonalization of A is then given by

$$U^\dagger A U = D,$$

where D is the diagonal matrix containing the eigenvalues of A.
Example: Consider again the matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Thus,

$$U = \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The matrix A becomes diagonal in this eigenbasis:

$$U^\top A U = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

2 Functional Calculus

The whole idea here is to extend a function $f : \mathbf{R} \to \mathbf{R}$ to matrices. A polynomial function can be directly extended and applied on matrices, non-polynomial functions need either series expansion or spectral theorem. Any Hermitian matrix H on a finite-dimensional Hilbert space can be diagonalized:

$$H = \sum_n \lambda_n P_n,$$

For any function $f : \mathbb{R} \to \mathbb{R}$ (or $\mathbb{C}$), the functional calculus defines

$$f(H) := \sum_n f(\lambda_n) P_n.$$

In the eigenbasis of H, which diagonalizes it, we have

$$f(H) = \mathrm{diag}(f(\lambda_1), f(\lambda_2), \ldots, f(\lambda_n)).$$

Example:

$$H = \begin{pmatrix} 2 & 3i \\ -3i & 2 \end{pmatrix}, \quad f(x) = e^x.$$

The eigenvectors and eigenvalues are,

$$\lambda_1 = 5 \quad |v_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \qquad \lambda_2 = -1 \quad |v_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

Using spectral theorem, we can write,

$$f(A) = f(\lambda_1)|v_1\rangle\langle v_1| + f(\lambda_2)|v_2\rangle\langle v_2|$$

$$f(A) = e^5|v_1\rangle\langle v_1| + e^1|v_2\rangle\langle v_2|$$

$$f(A) = e^2 \frac{1}{2} \begin{pmatrix} 1 \\ -i \end{pmatrix} \begin{pmatrix} 1 & i \end{pmatrix} + e \frac{1}{2} \begin{pmatrix} 1 \\ i \end{pmatrix} \begin{pmatrix} 1 & -i \end{pmatrix}$$

Which can be easily further manipulated to get a final matrix $f(A)$.
If an operator is a diagonal form already, then it is already in eigen basis and the function can be directly applied to the diagonal elements.

3 Axioms of Quantum Mechanics

1. **States:** The state of a quantum system is represented by a vector $|\psi\rangle$ in a complex Hilbert space $\mathcal{H}$. Two vectors that differ by a global phase represent the same physical state:

$$|\psi\rangle \sim e^{i\theta}|\psi\rangle, \quad \theta \in \mathbb{R}.$$

2. **Observables:** Every measurable quantity is represented by a Hermitian operator A on $\mathcal{H}$:

$$A = A^\dagger.$$

The expectation value is given by

$$\langle A \rangle = \sum a_i p_i = \sum a_i \langle \psi | P_i | \psi \rangle$$

3. **Measurement:** If A has eigenvalues $\{a_i\}$ and orthonormal eigenvectors $\{|a_i\rangle\}$, then a measurement of A in state $|\psi\rangle$ yields outcome a_i with probability

$$p_i = |\langle a_i|\psi\rangle|^2 = \langle\psi|P_i|\psi\rangle$$

First equality is the definition, second equality can be obtained though some simple calculations. After the measurement, the system collapses to $|a_i\rangle$ corresponding to the observed eigenvalue.

4. **Time Evolution:** The time evolution of a closed quantum system is unitary. If the system has Hamiltonian H, then

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle, \quad U(t) = e^{-iHt/\hbar}, \quad U^\dagger U = I.$$

5. **Composite Systems:** The state space of a composite system consisting of subsystems A and B is the tensor product of the subsystems' state spaces:

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B.$$

6. **Density Matrices (Mixed States):** For statistical mixtures, the state is represented by a density matrix ρ with properties:

$$\rho = \sum_i p_i|\psi_i\rangle\langle\psi_i|, \quad p_i \geq 0, \quad \sum_i p_i = 1.$$

$$\rho^\dagger = \rho, \quad \langle\psi|\rho|\psi\rangle \geq 0, \quad \mathrm{Tr}(\rho) = 1.$$

Expectation value of an observable A is

$$\langle A\rangle = \mathrm{Tr}(\rho A).$$

Probability of measuring a eigenvalue a,

$$p_a = \mathrm{Tr}(\rho P_a)$$

State just after measurement.

$$\rho_a' = \frac{P_a\rho P_a}{\mathrm{Tr}(\rho P_a)} = \frac{P_a\rho P_a}{p_a}$$

4 Mixed States: Density Matrix

A general physical system always exists in a mix of pure ψ states, thus we define the density matrix ρ, which even though represents a state, but is mathematically a hermitian operator on the Hilbert space $\mathcal{H}$. Additionally, $\mathrm{Tr}(\rho) = 1$ and diagonal elements are non-negative.
The set of all valid quantum states (pure or mixed) is denoted

$$\mathrm{DM}(\mathcal{H}).$$

It is defined by

$$\mathrm{DM}(\mathcal{H}) = \left\{\rho \in \mathcal{B}(\mathcal{H}) \,:\, \rho = \rho^\dagger,\ \rho \geq 0,\ \mathrm{Tr}(\rho) = 1\right\}.$$

Here $\mathcal{B}(\mathcal{H})$ is the set of all bounded linear operators on a Hilbert space $\mathcal{H}$.

A simple derivation and the context of the density matrix can be obtained through the following analysis. For a mixed state

$$\langle A\rangle = \sum_i p_i\langle\psi_i|A|\psi_i\rangle$$

Where p_i are classical probabilities to find the system in state ψ_i and $\langle\psi_i|A|\psi_i\rangle$ is the expectation value of the operator A in the given state ψ_i,

$$\langle A\rangle = \sum_i p_i\langle\psi_i|A|\sum_n |n\rangle\langle n|\psi_i\rangle$$

Since $|n\rangle$ form an orthonormal basis,

$$\langle A\rangle = \sum_i \sum_n p_i \langle\psi_i|A||n\rangle\langle n|\psi_i\rangle = \sum_i \sum_n p_i \langle n|\psi_i\rangle\langle\psi_i|A|n\rangle$$

$$\langle A\rangle = \sum_i p_i \mathrm{Tr}(|\psi_i\rangle\langle\psi_i|A)$$

$$\langle A\rangle = \mathrm{Tr}\left(\sum_i p_i(|\psi_i\rangle\langle\psi_i|A)\right)$$

We define the density matrix of mixed state as,

$$\rho = \sum_i p_i|\psi_i\rangle\langle\psi_i|$$

Then the expectation value becomes,

$$\langle A\rangle = \mathrm{Tr}(\rho A)$$

Further, the probability of measuring of its eigenvalue λ_i is given by,

$$p_i = \mathrm{Tr}(\rho P_i)$$

Where P_i is the projector operator corresponding to that eigenvalue, obtained by using the corresponding eigenvectors $|a_i\rangle\langle a_i|$.
The state of the system immediately after measurement such that the measured value is some eigenvalue λ_i will be,

$$\rho_i = \frac{P_i \rho P_i}{\mathrm{Tr}(\rho P_i)}$$

The conditions have the following meanings:
Hermiticity:

$$\rho = \rho^\dagger.$$

Ensures all expectation values $\mathrm{Tr}(\rho A)$ of Hermitian observables A are real.
Positivity:

$$\rho \geq 0 \quad \Longleftrightarrow \quad \langle\psi|\rho|\psi\rangle \geq 0 \quad \forall\, |\psi\rangle \in \mathcal{H}.$$

Since, ρ is hermitian, we only need to show that $\det(\rho) \geq 0$ and $\rho_{11} \geq 0$. Guarantees probabilities are non-negative.
Normalization:

$$\mathrm{Tr}(\rho) = 1.$$

Ensures total probability equals 1.

5 Gibbs State and Entropy

A Gibbs state ρ_B is a state consistent with the given temperature T,

$$\rho = \frac{e^{-\beta H}}{Z}$$

Where H is the Hamiltonian of the system and $Z = \mathrm{Tr}(e^{-\beta H})$ is the partition function. Using functional calculus,

$$\rho = \frac{\sum \exp(\beta E_n)|n\rangle\langle n|}{\mathrm{Tr}(\sum \exp(\beta E_n)|n\rangle\langle n|)}$$

$$\rho = \sum p_i |n\rangle\langle n|$$

Where p_i are defined as ,

$$p_i = \frac{e^{-\beta E_n}}{Z}$$

For each state ρ, we define its entropy by,

$$S(\rho) = -\mathrm{Tr}(\rho \ln \rho) =$$

Few properties:

- Non-negativity $S(\rho) \geq 0$.
- For pure state $\rho = |\psi\rangle\langle\psi|$, entropy $S(\rho) = 0$. For mixed states $S(\rho) > 0$.

Using spectral theorem, we can write,

$$S(\rho) = -\mathrm{Tr}(\sum \lambda_i P_i \sum \ln(\lambda_i) P_i) = -\mathrm{Tr}(\sum \lambda_i \ln(\lambda_i) P_i) = \sum \lambda_i \ln(1/\lambda_i)$$

Where λ_i are the eigenvalues of the density matrix ρ.

Maximisation of Entropy: We consider the maximization problem of $S(\rho)$ under the constraint $\sum \lambda = 1$(Since $\mathrm{Tr}(\rho) = 1$), we consider the lagrangian,

$$L = -\sum \lambda_i \ln(\lambda_i) + \alpha \left(\sum \lambda_i - 1 \right)$$

Differentiating with respect to λ and equating to zero,

$$0 = -\sum (1 + \ln(\lambda_i) + \alpha \sum_i 1 = 0$$

$$1 + \ln \lambda_i + \alpha = 0$$

$$\implies \lambda_i = \text{constant}$$

From the normalization property, we get $\lambda_i = \frac{1}{dim(H)}$ Thus the density matrix or the mixed state which maximises the entropy is

$$\rho_{max} = \frac{1}{d} I$$

$$S_{max} = \ln(d)$$

Entropy maximization for fixed energy: Gibbs state maximizes the entropy for a fixed given total energy. Gibbs state ρ_B maximizes the entropy.
Relative entropy of ρ with respect to σ is defined by,

$$S(\rho|\sigma) = \mathrm{Tr}(\rho \ln \rho - \rho \ln \sigma)$$

We have $S(\rho|\sigma) \geq 0$ and $S(\rho|\sigma) = 0 \quad \text{Iff} \quad \rho = \sigma$.

6 Convexity and Concavity

Convex combinations

Let x, y belong to a vector space and let $t \in [0, 1]$. An expression of the form

$$tx + (1 - t)y$$

is called a *convex combination* of x and y.

Convex sets

A set C is said to be *convex* if it is closed under convex combinations, i.e.

$$\forall x, y \in C,\ \forall t \in [0,1], \quad tx + (1-t)y \in C.$$

Geometrically, this means that the line segment joining any two points of C lies entirely inside C.

Convex and concave functions

Let f be a real-valued function defined on a convex set.

Convex function. The function f is *convex* if

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y), \quad \forall x, y,\ \forall t \in [0,1].$$

Equivalently, the graph of f lies below the straight line joining the points $(x, f(x))$ and $(y, f(y))$.

Concave function. The function f is *concave* if

$$f(tx + (1-t)y) \geq tf(x) + (1-t)f(y), \quad \forall x, y,\ \forall t \in [0,1].$$

Equivalently, the graph of f lies above the straight line joining the points $(x, f(x))$ and $(y, f(y))$.

Convexity of the set of density matrices

Let $\mathcal{H}$ be a finite-dimensional Hilbert space and denote by $DM(\mathcal{H})$ the set of density matrices on $\mathcal{H}$. Recall that $\rho \in DM(\mathcal{H})$ if and only if:

- $\rho^\dagger = \rho$,
- $\rho \geq 0$,
- $\operatorname{Tr} \rho = 1$.

Proposition. The set $DM(\mathcal{H})$ is convex.

Proof. Let $\rho_1, \rho_2 \in DM(\mathcal{H})$ and $t \in [0,1]$. Define

$$\rho(t) = t\rho_1 + (1-t)\rho_2.$$

Then:

$$\begin{aligned}
\rho(t)^\dagger &= t\rho_1^\dagger + (1-t)\rho_2^\dagger = \rho(t), \\
\langle\psi|\rho(t)|\psi\rangle &= t\langle\psi|\rho_1|\psi\rangle + (1-t)\langle\psi|\rho_2|\psi\rangle \geq 0, \\
\operatorname{Tr} \rho(t) &= t \operatorname{Tr} \rho_1 + (1-t) \operatorname{Tr} \rho_2 = 1.
\end{aligned}$$

Hence $\rho(t) \in DM(\mathcal{H})$. $\square$

Concavity of the von Neumann entropy

The von Neumann entropy of a density matrix ρ is defined by

$$S(\rho) = -\operatorname{Tr}(\rho \ln \rho).$$

Property. The von Neumann entropy is concave on $DM(\mathcal{H})$, i.e.

$$S\big(t\rho_1 + (1-t)\rho_2\big) \geq tS(\rho_1) + (1-t)S(\rho_2), \quad \forall t \in [0,1].$$

Consequences

From the convexity of $DM(\mathcal{H})$ and the concavity of S, we obtain:

- $S(\rho) \geq 0$, with $S(\rho) = 0$ if and only if ρ is pure;
- $S(\rho) \leq \ln(\dim \mathcal{H})$;
- the maximum entropy is attained for the maximally mixed state

$$\rho = \frac{I}{\dim \mathcal{H}}.$$

7 Quantum Trajectories

The basic idea of quantum trajectories is we start with a state ρ and we measure some operator $A = \sum a_i P_i$ whose expectation value is $\mathrm{Tr}(\rho A)$. The outcomes comes out to be a_i with probability $p_i = \mathrm{Tr}(\rho P_i)$. After this we let the system evolve under unitary time evolution and then again measure A and consider the conditional probabilities.

Let

$$A = \sum_i a_i P_i$$

be an observable. According to the rules of quantum mechanics, measuring A yields the outcome a_i with probability

$$p_i = \mathrm{Tr}(P_i\rho) = \mathrm{Tr}(\rho P_i) = \mathrm{Tr}(P_i \rho P_i).$$

Instantaneously after the measurement, the state collapses to

$$\rho \longmapsto \frac{P_i \rho P_i}{\mathrm{Tr}(P_i \rho P_i)}, \qquad \text{if } \mathrm{Tr}(P_i \rho P_i) > 0.$$

The expectation value of the outcome of the measurement of A is

$$\langle A \rangle = \sum_i a_i p_i = \sum_i a_i \mathrm{Tr}(P_i \rho) = \mathrm{Tr}(A\rho)$$

Moreover, the expectation value of post-measurement states will be,

$$\left\langle \frac{P_i \rho P_i}{\mathrm{Tr}(P_i \rho P_i)} \right\rangle = \sum_i \mathrm{Tr}(\rho P_i) \frac{P_i \rho P_i}{\mathrm{Tr}(P_i \rho P_i)} = \sum_i P_i \rho P_i.$$

This represents post measurement, un selective state. Physically, it means we forgot the outcome of first measurement.

7.1 Two successive measurements

Quantum Joint Probability Interpretation:

In quantum mechanics, the joint probability

$$p(a_i, a_j) = \mathrm{Tr}(P_j U P_i \rho P_i U^*)$$

describes the probability of a *sequential process*: first measuring a_i, then evolving under U, and finally measuring a_j. It is not the probability of observing both outcomes simultaneously.
This still satisfies the conditional probability rule:

$$p(a_i, a_j) = p(a_i) \cdot p(a_j \mid a_i),$$

but "a_i and a_j" means a measurement trajectory $a_i \to a_j$, not simultaneous truth as in classical probability.

After obtaining the result a_i at the first measurement, the state evolves as

$$\rho_i' = \frac{U P_i \rho P_i U^*}{\mathrm{Tr}(P_i \rho P_i)}.$$

This state encodes the fact that **we have measured A and found a_i to be the outcome and then let the state evolve by unitary evolution**.
Measuring A again now, the probablity to get a_j on the measurement of A will use conditional probability

$$p(a_j \mid a_i) = \mathrm{Tr}(\rho_i' P_j) = \mathrm{Tr}\left(P_j \frac{U P_i \rho P_i U^*}{\mathrm{Tr}(P_i \rho P_i)}\right) = \frac{\mathrm{Tr}(P_j U P_i \rho P_i U^*)}{\mathrm{Tr}(P_i \rho P_i)}.$$

By conditional probability,

$$p(a_i, a_j) = p(a_i)\, p(a_j \mid a_i) = \mathrm{Tr}(P_j U P_i \rho P_i U^*).$$

We also have

$$\sum_{i,j} p(a_i, a_j) = \sum_{i,j} \mathrm{Tr}(P_j U P_i \rho P_i U^*) = \sum_i \mathrm{Tr}\left(\sum_j P_j U P_i \rho P_i U^*\right) = \sum_i \mathrm{Tr}(U P_i \rho P_i U^*) = \sum_i = \mathrm{Tr}(\rho) = 1$$

Also, from usual probability rules we have,

$$\sum_i p(a_i, a_j) = 1$$

$$\rho \xrightarrow{\text{measure } A} \begin{cases} \rho_1 = \dfrac{P_1 \rho P_1}{p_1} & \text{with prob. } p_1 \\ \rho_2 = \dfrac{P_2 \rho P_2}{p_2} & \text{with prob. } p_2 \\ \vdots \end{cases}$$

$$\text{forget outcome} \quad \Longrightarrow \quad \rho_{\text{post}} = \sum_i p_i \rho_i = \sum_i P_i \rho P_i$$

$\boxed{\textbf{Coherent path (no intermediate measurement)}}$ $\rho \xrightarrow{U} U \rho U^\dagger \;\to\; p(a_j) = \mathrm{Tr}\big(P_j\, U \rho U^\dagger\big),\; S(\rho) = S(U \rho U^\dagger)$

$\boxed{\textbf{Decohered path (measure at } t_1 \textbf{ and forget outcome)}}$ $\rho \xrightarrow{\text{measure and forgot } A} \sum_i P_i \rho P_i \xrightarrow{U} U\Big(\sum_i P_i \rho P_i\Big) U^\dagger$

$$\implies \quad \tilde{p}(a_j) = \mathrm{Tr}\Big(P_j\, U(\sum_i P_i \rho P_i)U^\dagger\Big),\ S\Big(\sum_i P_i \rho P_i\Big) \geq S(\rho).$$

What if you don't care about which result you got in the first measurement?
You just want the probability of getting outcome a_j in the second measurement. This is called marginalization over the first measurement. This can be obtained in two equivalent ways.

Probability of getting a_i from non-selective post measurement state = Total probability of getting a_j after the first outcome, summed over all first outcomes. $$\mathrm{Tr}(P_j U \sum_i (P_i \rho P_i) U^\dagger P_j) = \sum_i p(a_i, a_j)$$

The law of the second measurement is obtained by marginalisation:

$$\widetilde{p}(a_j) = \sum_i p(a_i, a_j) = \sum_i \mathrm{Tr}(P_j U P_i \rho P_i U^*) = \mathrm{Tr}\left(P_j U\Big(\sum_i P_i \rho P_i\Big)U^*\right).$$

This includes the effect of the first measurement (decoherence), but not the knowledge of which outcome happened. .

Compare with the distribution without first measurement:

$$p(a_j) = \mathrm{Tr}(P_j U \rho U^*).$$

This difference expresses the *decoherence effect* of the first measurement.

After performing the observation of A on the system, but not recording the measurement, the states becomes

$$\rho_{decohered} = \sum P_i \rho P_i$$

Compare this to usual post measurement state, where we do record the measurement,

$$\rho = \frac{P_i \rho P_i}{\mathrm{Tr}(P_i \rho P_i)}$$

Define

$$\mathcal{D}(\rho) := \sum_i P_i \rho P_i.$$

Going from a usual mixed state to a post measurement unrecorded state, the entropy increases, because we are destroying the information.

$$S(D(\rho)) \geq S(\rho)$$

Considering the relative entropy,

$$S(\rho|D(\rho)) = \mathrm{Tr}(\rho \ln \rho) - \mathrm{Tr}(\rho \ln D(\rho))$$

We have,

$$\mathrm{Tr}\rho \ln D(\rho) = \mathrm{Tr}\rho \ln \left(\sum_i P_i \rho P_i\right)$$

Since $S(\rho|D(\rho)) \geq 0$, this implies,

$$S(D(\rho)) \geq S(\rho) \geq S(\rho_i)$$

Furthermore, also $\Delta S = S(\rho|D(\rho))$. From launders bound, any entropy change process must lead to a heat dissipation to the environment given by,

$$\Delta Q_{min} \geq K_B T \Delta S_{erased}$$

8 Variation of the measured observable

Let A be an observable with spectral decomposition

$$A = \sum_i a_i P_i,$$

where $\{P_i\}$ are orthogonal projectors, and let the system initially be in a (possibly mixed) state ρ.

At time $t = 0$, a projective measurement of A is performed. If the outcome a_i is obtained, the (selective) post-measurement state is

$$\rho_i = \frac{P_i \rho P_i}{\mathrm{Tr}(P_i \rho)}, \qquad \text{with probability } \mathrm{Tr}(P_i \rho).$$

If the outcome is not recorded (non-selective measurement), the post-measurement state is given by the Lüders map

$$\boxed{\mathcal{D}(\rho) = \sum_i P_i \rho P_i.}$$

Important: $\mathcal{D}(\rho)$ represents the state *after the first measurement only*, averaged over all possible outcomes a_i. It does not involve the second measurement.

After the first measurement, the system evolves for a time T under a unitary operator U, so that

$$\mathcal{D}(\rho) \;\longrightarrow\; U\,\mathcal{D}(\rho)\,U^*.$$

At time $t = T$, the observable A is measured again. The joint probability of obtaining outcomes (a_i, a_j) in the two measurements is

$$p(a_i, a_j) = \mathrm{Tr}(P_j\, U\, P_i \rho P_i\, U^*)\,.$$

Define the variation of the observable as

$$\delta = a_j - a_i.$$

The probability distribution of δ is

$$p(\delta) = \sum_{i,j:\, a_j - a_i = \delta} p(a_i, a_j).$$

The expectation value of the variation is therefore

$$\mathbb{E}(\delta) = \sum_{i,j} (a_j - a_i)\, \mathrm{Tr}(P_j\, U\, P_i \rho P_i\, U)\,.$$

Using $\sum_j a_j P_j = A$ and $\sum_j P_j = \mathbb{I}$, one obtains

$$\mathbb{E}(\delta) = \mathrm{Tr}(A\, U\, \mathcal{D}(\rho)\, U) - \mathrm{Tr}(A\rho).$$

Equivalently,

$$\boxed{\mathbb{E}(\delta) = \langle A \rangle_{U\mathcal{D}(\rho)U - \langle A \rangle_\rho}.}$$

Key point: The two-point measurement scheme does not probe the change of A under the unitary evolution $\rho \mapsto U\rho U^*$, but rather under the measurement-induced process

$$\rho \;\longrightarrow\; U\,\mathcal{D}(\rho)\,U^*,$$

since the first projective measurement irreversibly dephases the state in the eigenbasis of A.

In the special case $U = \mathbb{I}$, the expectation value of the variation vanishes,

$$\mathbb{E}(\delta) = 0,$$

even though individual measurement outcomes may fluctuate.

9 Multiple measurements and trajectories

For successive measurements,

$$\mathbb{P}_\rho^A(a_{i_1}, \ldots, a_{i_n}) = \mathrm{Tr}(P_{i_n} U \cdots P_{i_1} \rho P_{i_1} \cdots U^* P_{i_n}).$$

In the pure state case $\rho = |\psi\rangle\langle\psi|$,

$$\mathbb{P}_\rho^A(a_{i_1}, \ldots, a_{i_n}) = \prod_{k=1}^{n} |\langle e_{i_k}|U|e_{i_{k-1}}\rangle|^2,$$

yielding a Markov chain with transition probabilities

$$\pi_{ij} = |\langle e_j|U|e_i\rangle|^2.$$

10 Bi-partite systems

Definitions and Concepts

Bipartite systems consist of two subsystems, A and B, each with its own Hilbert space. The total Hilbert space for the composite system is the tensor product of the subsystems' Hilbert spaces.

Let $\mathcal{H}_A$ and $\mathcal{H}_B$ be Hilbert spaces of dimensions d_A and d_B, respectively. The composite system is described by the tensor product space:

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B,$$

with the dimension of $\mathcal{H}_{AB}$ being $d_A \cdot d_B$.

Basis: If $\{e_i\}_{i=1}^{d_A}$ is an orthonormal basis of $\mathcal{H}_A$ and $\{s_j\}_{j=1}^{d_B}$ is an orthonormal basis of $\mathcal{H}_B$, then the canonical basis of the joint system is given by:

$$\{e_i \otimes s_j \mid i = 1, \ldots, d_A; \quad j = 1, \ldots, d_B\}.$$

Any vector $|\psi\rangle \in \mathcal{H}_{AB}$ can be expressed as a linear combination of these basis vectors:

$$|\psi\rangle = \sum_{i=1}^{d_A} \sum_{j=1}^{d_B} c_{ij}\, e_i \otimes s_j$$

where c_{ij} are complex coefficients.

Operators on Bipartite Systems: If $X \in B(\mathcal{H}_A)$ and $Y \in B(\mathcal{H}_B)$ are operators acting on subsystems A and B, respectively, then the combined operator on the composite system is given by the tensor product:

$$X \otimes Y.$$

The individual operators act on the product state as following,

$$X|\psi\rangle \otimes |\phi\rangle = (X \otimes I_B)(|\psi\rangle \otimes |\phi\rangle)$$

$$X|\psi\rangle \otimes |\phi\rangle = (X|\psi\rangle) \otimes |\phi\rangle$$

Similarly with Y; $Y(|\psi\rangle \otimes |\phi\rangle) = |\psi\rangle \otimes (Y|\phi\rangle)$

Important!!: How operators from subsystems act on composite system:
The act of applying a operator from one subsystem on the composite state firstly involves extending the operator to other subsystem through $X \otimes I_B$ or $I_A \otimes Y$, then the subsequent action is given by $(X|\psi\rangle) \otimes |\phi\rangle$ and $|\psi\rangle \otimes (Y|\phi\rangle)$

Some important and useful identities:

Adjoint operation is distributable over tensor products: $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$

Also, $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$

A more basic identity, $(ABCD\ ...Z)^\dagger = Z^\dagger\ ...D^\dagger C^\dagger B^\dagger A^\dagger$

$\mathrm{Tr}(ABC) = \mathrm{Tr}(BCA) = \mathrm{Tr}(CAB) \quad \mathrm{Tr}(A \otimes B \otimes C\ ...) = \mathrm{Tr}(A)\mathrm{Tr}(B)\mathrm{Tr}(C)\ ...$

The action of $X \otimes Y$ on a product state is defined as:

$$(X \otimes Y)(|a\rangle \otimes |b\rangle) = (X|a\rangle) \otimes (Y|b\rangle).$$

Partial Trace and Reduced Density Matrices

Definition. Given a composite system with Hilbert space

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B,$$

The *partial trace over subsystem* B is the linear map

$$\mathrm{Tr}_B : B(\mathcal{H}_{AB}) \to B(\mathcal{H}_A)$$

Such that, it removes or traces out the B part from the composite system and provides the system A.

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}).$$

If A and B are operators acting on the individual subsystems, then,

$$\mathrm{Tr}_A(A \otimes B) = \mathrm{Tr}(A)B \quad \text{and} \quad \mathrm{Tr}_B(A \otimes B) = A\mathrm{Tr}(B)$$

Interpretation The partial trace extracts "everything that is observable" from the viewpoint of one subsystem.
Operationally, given a density matrix ρ in composite basis $(i,j)(k,l)$, we can find the partial traces using,

$$\mathrm{Tr}_A(\rho)_{jl} = \rho_{B(jl)} = \sum_i \rho_{(i,k),(i,l)}$$

$$\mathrm{Tr}_B(\rho)_{ik} = \rho_{A(ik)} = \sum_j \rho_{(i,j)(k,j)}$$

The entropy of composite $\rho_A \otimes \rho_B$ is given by,

$$S(\rho_A \otimes \rho_B) = S(\rho_A) + S(\rho_B)$$

$$S(\rho_{AB}) \leq S(\rho_A) + S(\rho_B)$$

11 Dynamics of Open Quantum Systems

Consider a system $\mathcal{H}_S$ and environment $\mathcal{H}_e$, their coupled dynamics is decribed by,

$$\mathcal{H}_\lambda = \mathcal{H}_A \otimes \mathcal{I}_e + \mathcal{I}_A \otimes \mathcal{H}_e + \lambda\omega$$

Given the initial state of the full system $\rho_S(o) \otimes \sigma$, where $\rho_s(0) \in DM(H_S)$ and $\sigma \in DM(H_e)$.
The effective/reduced state on S at time t is given by,

$$\rho_S(t) = \mathrm{Tr}_e U(t)\rho_s(0) \otimes \sigma U^*(t)$$

We consider the linbald approximation, given by,

$$\frac{d}{dt}\rho(t) = \mathcal{L}(\rho(t))$$

The solution is given by,

$$\rho(t) = e^{t\mathcal{L}}\rho_0 = \sum_{n\geq 0} \frac{t^n \mathcal{L}^n}{n!}\rho_0$$

The linbald operator is given by,

$$\mathcal{L} = -i[H_s, \rho] + \sum_i (V_i \rho_i V_i^* - \frac{1}{2}\{V_i^* V_i, \rho\})$$